Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AS742]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

AEROSPACE AVIONICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain the different multiplexing techniques used in communication and its application.	**07**
	b.	Compare the operation of TCAS and ADS- B system.	**07**
2.	**a.**	Explain satellite communication with block diagram.	**10**
	b.	Calculate the maximum range of Radar for given specifications. Pt = 500 KW, G = 6000 , Ae = 6 m2 , σ = 40 m2, Smin = 10^{-12} W.	**04**
3.	**a.**	Explain the curved Earth Distance and estimate the curved Earth distance from A to B with the following values ΦA=25° N , ΦB = 82° N; A = 75°E, B= 55°W, Earth radius = 6370 Km.	**07**
	b.	Explain the operation of LORAN C system.	**07**
		OR	
4.	**a.**	Explain the operation of Automatic Direction Finder used in Radio Navigation	**07**
	b.	What is the length of a 1° increment along a meridian and on a parallel at 40N, 80W? Radius of the earth = 6370 km.	**07**
5.	**a.**	Compare the ARNIC 429 and MIL 1553 B BUS communication system	**07**
	b.	Explain the operation of AFDX network in an aircraft system .	**07**
		OR	
6.	**a.**	Explain with block diagram integrated modular Avionics system.	**07**
	b.	Explain the different sources errors and its effect on GPS signal.	**07**
7.	**a.**	Explain how do you get the position using GPS Triangulation.	**07**
	b.	Explain the GPS architecture and computation of DOP and velocity .	**07**
		OR	
8.	**a.**	Discuss the application of GPS for surface vehicles, atmospheric flight vehicles and satellites.	**07**
	b.	Describe the requirements and operation of differential GPS.	**07**